

PE-EC703C — Wavelet Transforms

Semester VII • 3L:0T:0P • 3 Credits

Module 7 Exam Notes: Beyond Wavelet

CGEC | B.Tech ECE (2023–27) | *Biraj Sarkar*

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1. Limitations of Wavelets in Two Dimensions

Standard 2D wavelets are created by tensor products of 1D wavelets (separable DWT). Their spatial support is square (isotropic). While excellent at representing point singularities, they struggle to capture **line or curve singularities** (such as edges in an image).

To represent a smooth curve, 2D wavelets require a large number of coefficients at fine scales, failing to achieve optimal **sparsity**. This limitation is known as the *curse of dimensionality* for directional features.

2. Ridgelet Transform

The continuous Ridgelet transform is designed specifically to capture line singularities in 2D. A ridgelet function in $\mathbb{R}^2$ is defined by scaling (a), translating (b), and rotating (θ):

$$\psi_{a,b,\theta}(x,y) = a^{-1/2} \psi\left(\frac{x \cos \theta + y \sin \theta - b}{a}\right)$$

Here, the wavelets are constant along the lines $x \cos \theta + y \sin \theta = \text{constant}$ (ridges).

Relationship to Radon Transform: The Ridgelet transform can be implemented by first taking the **Radon Transform** of the image $f(x,y)$ (which maps line integrals to points) and then applying a **1D Wavelet Transform** along the radial direction:

$$\text{Ridgelet Transform} = \text{1D DWT} \circ \text{Radon Transform}$$

3. Curvelet Transform

While Ridgelets handle straight line singularities, real-world images contain curved edges. The **Curvelet Transform** generalizes Ridgelets to handle curves by partitioning the image into small local patches, and applying a local Ridgelet transform to each patch.

3.1 Parabolic Scaling of Curvelets

A key mathematical property of curvelets is **Parabolic Scaling**, which governs the relationship between the width and length of their spatial support at scale j :

Parabolic Scaling Law

$$\text{width} \approx \text{length}^2 \quad \Rightarrow \quad \text{width} \propto 2^{-j}, \quad \text{length} \propto 2^{-j/2}$$

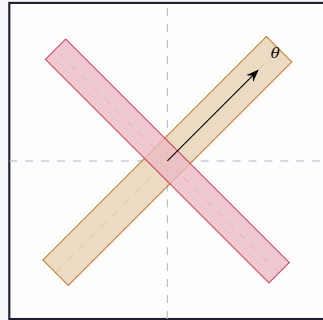
At fine scales, curvelets become highly needle-like (highly anisotropic), allowing them to align perfectly with curved edge structures.

3.2 Digital Curvelet Transform Implementation

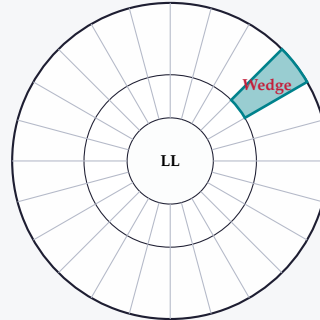
Two fast algorithms exist for the discrete curvelet transform:

1. **Unequally Spaced Fast Fourier Transform (USFFT):** Samples the Fourier transform of the image along a polar-like grid of wedges.
2. **Wrapping Algorithm:** Wraps Fourier wedges into a rectangular grid around the origin, which is faster and computationally more efficient.

Ridgelet vs. Curvelet Frequency Domain Partitioning



(a) Ridgelet: Radial Slabs



(b) Curvelet: Polar Wedges

4. Properties and Applications of Ridgelets and Curvelets

• Properties:

- **High Directional Sensitivity:** Support orientations increase at finer scales.
- **Optimal Sparsity:** Curvelets represent curved edges with very few non-zero coefficients compared to wavelets.

• Applications:

- **Seismic Data Processing:** Identifying straight and curved geological faults.
- **CT/MRI Reconstruction:** Solving tomographic reconstruction problems from radon projections.
- **Directional Image Denoising:** Removing noise while preserving sharp curved edges.

Quick Revision — Module 7

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Key Equations

Ridgelet Transform $\psi_{a,b,\theta}(x, y) = a^{-1/2} \psi\left(\frac{x \cos \theta + y \sin \theta - b}{a}\right)$

Parabolic Scaling Law $\text{width} \approx \text{length}^2 \implies \text{width} \propto 2^{-j}, \text{length} \propto 2^{-j/2}$

One-Line Definitions

- **Anisotropic:** Having properties that differ depending on the direction of measurement (e.g., curvelet shape).
- **Radon Transform:** Mathematical integral projection mapping line integrals of a 2D function into 1D profiles.
- **Ridgelet Transform:** Directional 2D transform mapping straight line singularities to point-like representations.
- **Curvelet Transform:** Directional multiscale transform optimized for curved edge singularities.
- **Wrapping Algorithm:** Fast discrete curvelet transform that wraps Fourier wedges into a rectangular grid.
- **USFFT Algorithm:** Fast discrete curvelet transform that uses unequally spaced fast Fourier projections.

Frequently Asked / Viva Questions

Q1. Why are standard wavelets inefficient at representing curved edges in 2D?

Separable 2D wavelets have square spatial support. Representing a smooth curved edge requires many small wavelets along the curve at fine scales, which destroys sparsity.

Q2. How is the Ridgelet transform implemented using the Radon transform?

The Ridgelet transform is implemented by first applying the Radon transform to project line integrations into 1D profiles, followed by a 1D continuous wavelet transform along those radial profiles.

Q3. What is the parabolic scaling property of curvelets?

Curvelets scale such that $\text{width} \approx \text{length}^2$ (i.e. at scale 2^{-j} , the width is 2^{-j} and length is $2^{-j/2}$). This anisotropic, needle-like shape allows curvelets to trace curved edges efficiently.

Q4. Explain the two primary digital curvelet transform algorithms.

Frequently Asked / Viva Questions (contd.)

- **USFFT:** Computes spatial coordinates on an irregular polar-like grid in the frequency domain.
- **Wrapping:** Wraps rectangular frequency wedges into a standard rectangular grid to allow fast FFT processing.

Q5. What are the primary applications of Ridgelets and Curvelets?

They are used in seismic data processing, CT/MRI tomographic reconstruction, and denoising of images with dominant curved structures, where wavelets would blur the edges.

Multi-Resolution Feature Representation Comparison

Property	Wavelet Transform	Ridgelet Transform	Curvelet Transform
Dimension	1D, 2D, or Multi-D.	2D (or higher).	2D (or higher).
Optimal For	Point singularities.	Straight line singularities.	Curved singularities (edges).
Support Geometry	Isotropic (square).	Line-like (infinite slabs).	Needle-like (parabolic wedges).
Directionality	3 fixed directions in 2D.	Continuous angle θ .	Multi-directional (scale-dependent).
Fourier Tiling	Concentric dyadic squares.	Radial strips.	Polar-like wedges.